

AI Museum of the Future

Week 1 — Use Case, Scope, Team Roles, Automation Tool & First Architecture

1. Use Case Defined & Scoped

Core idea: Build an AI-powered virtual museum that represents our current era as if it were being curated 100 years in the future. The system automatically collects public contemporary data, analyzes it through several specialized AI agents, connects the findings with a knowledge base, and creates a coherent virtual exhibition.

Central question

“If a museum were created in 100 years to represent our time, what would it preserve, and how would it tell our story?”

INPUT	PROCESSING	OUTPUT
• News & contemporary events • Images & media • Science & research • Technology & innovation • Social & cultural trends	• Data collection & cleaning • RAG retrieval • Multimodal image analysis • Specialized agents • Cross-agent synthesis • Curatorial selection	• Exhibition title & concept • 4-6 museum sections • Selected items / artifacts • Context & descriptions • Narrative / storytelling • Visual presentation

Scope for the 5-week project

IN SCOPE	OUT OF SCOPE
• A controlled corpus from accessible public sources • Contemporary news, science, technology, culture and images • 4-5 specialized agents + curator • RAG knowledge base • Image understanding as the multimodal component • Automated end-to-end workflow with n8n • Final virtual exhibition with several sections	• Crawling the entire Internet • Predicting with certainty what future historians will value • Building a full 3D/VR museum • Training a foundation model from scratch • Handling an unlimited number of data sources

2. Team Roles

Role	Main responsibility
Data & APIs	Collect, clean, normalize and structure public data from selected sources.
RAG Engineer	Build the corpus, chunking strategy, embeddings, vector database and retrieval evaluation.
Multi-Agent Engineer	Design agent responsibilities, communication, orchestration and state/information flow.
Multimodal / Vision	Process images, extract visual information and integrate image analysis with the textual pipeline.
Prompt & Evaluation	Maintain the versioned prompt library, test prompts, document failures and define evaluation cases.
n8n & Integration	Build the end-to-end automation workflow, connect APIs/agents/tools and handle errors.

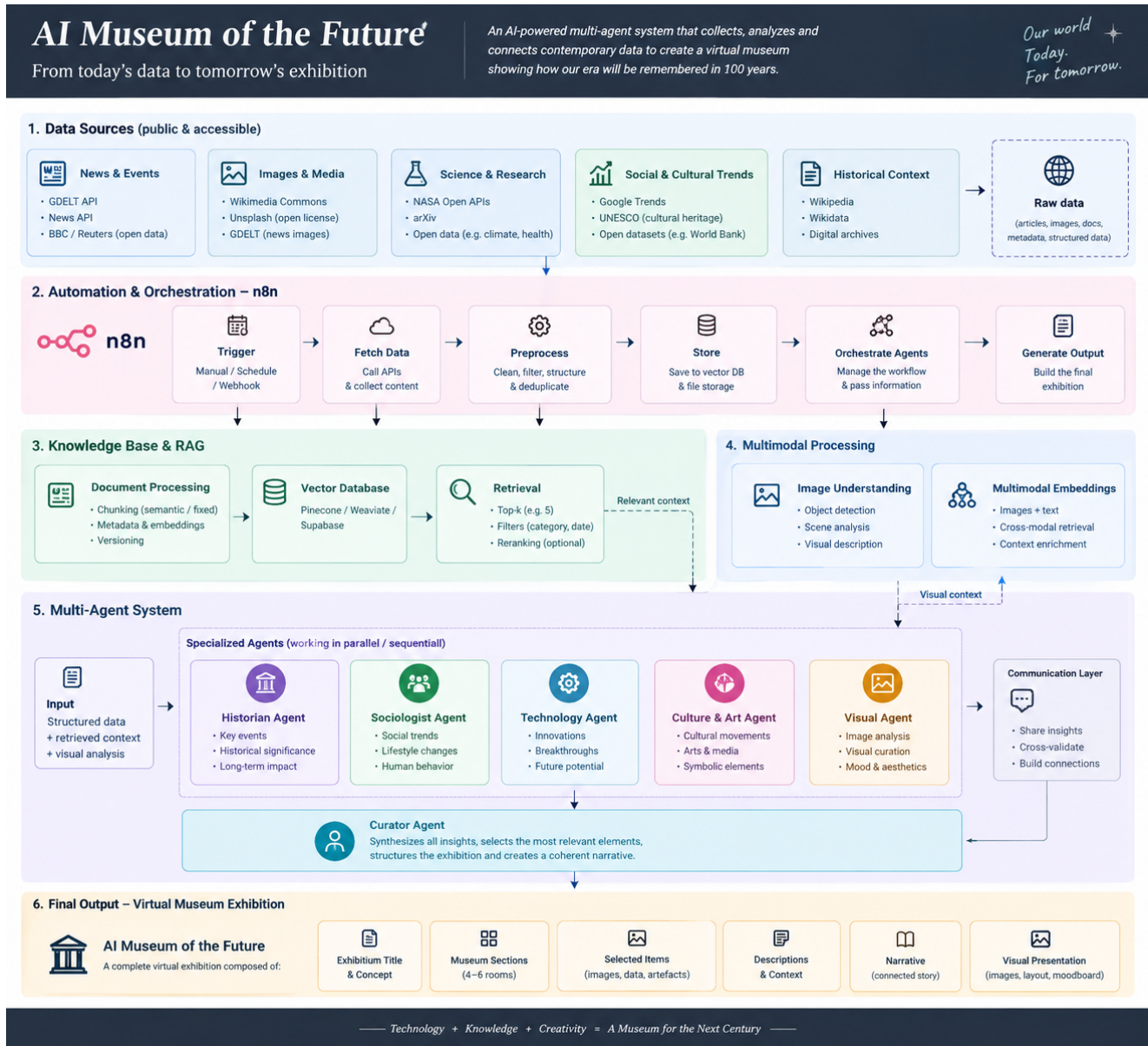
Team principle: these are responsibility areas rather than isolated silos. Every member should understand the complete pipeline and contribute to integration and testing.

3. Automation Tool Chosen

n8n is selected as the automation and orchestration backbone. It will trigger the workflow, collect data from APIs, route information between processing steps and agents, manage outputs, and make the entire pipeline reproducible without manual cell-by-cell execution.

Trigger options: manual trigger, webhook, or scheduled execution.

4. First Draft of the Architecture



Architecture logic

1. Public data layer: collect contemporary textual, visual and structured information.
2. n8n orchestration: trigger, fetch, preprocess, store and route information.
3. Knowledge layer: transform documents into a searchable knowledge base and retrieve relevant context with RAG.
4. Multimodal layer: analyze images and combine visual information with text and metadata.
5. Specialized agents: examine the same evidence from historical, social, technological, cultural and visual perspectives.
6. Curator Agent: synthesizes the agents' findings, selects coherent material and structures the exhibition.
7. Final output: generate a virtual exhibition with sections, selected items, context and a connected narrative.

Why this architecture? The project separates data acquisition, knowledge retrieval, multimodal understanding and creative synthesis. This makes each component testable while giving the multi-agent system a meaningful reason to exist: different agents provide complementary perspectives before the curator produces the final exhibition.